Data Analyst Capstone Project: Technology Trends Analysis

Lau Xian Jin

20 April 2026

OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY

- Core technologies (JavaScript, SQL, Python) dominate both current usage and future demand, indicating a stable technology foundation
- PostgreSQL leads database adoption, reinforcing the continued importance of relational data systems
- Job market analysis reveals uneven demand distribution and misalignment between demand and salary, highlighting complex market dynamics
- Effective career development requires balancing foundational skills for employability with specialized skills for differentiation
- The future technology landscape will be hybrid, combining established tools with emerging technologies such as cloud platforms and high-performance systems

INTRODUCTION

Purpose:

- ✓ Analyze global technology trends in programming, databases, and job demand

Target Audience:

- ✓ Aspiring data analysts
- ✓ Developers
- ✓ Hiring stakeholders

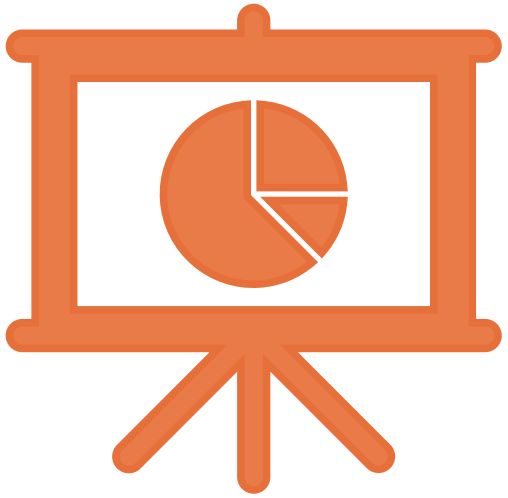
Value:

- ➢ Identify high-demand skills
- ➢ Understand future technology direction
- ➢ Support career decision-making

METHODOLOGY

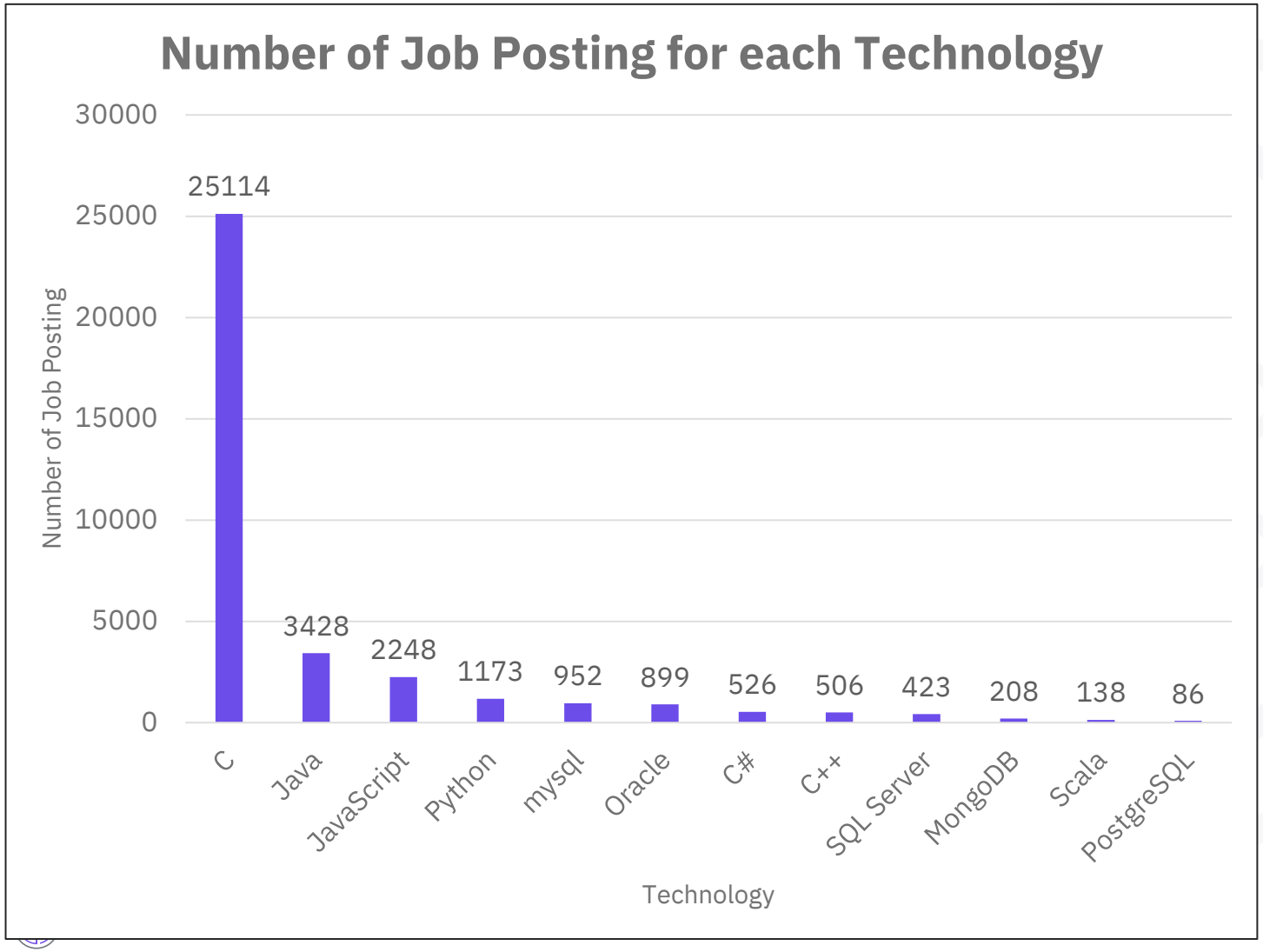
- Data Sources:
  - Stack Overflow Developer Survey
  - Job postings dataset (job-postings.xlsx)
- Data Processing:
  - Data cleaning (handling nulls, formatting)
  - Splitting multi-value fields (e.g. languages, databases)
  - Exploding columns for analysis
- Analysis Techniques:
  - Aggregation (groupby, counts)
  - Ranking (Top 10 technologies)
  - Visualization (bar charts, dashboards)

Results



The following slides present key findings from programming languages, databases, job market trends, and salary analysis.

JOB POSTINGS



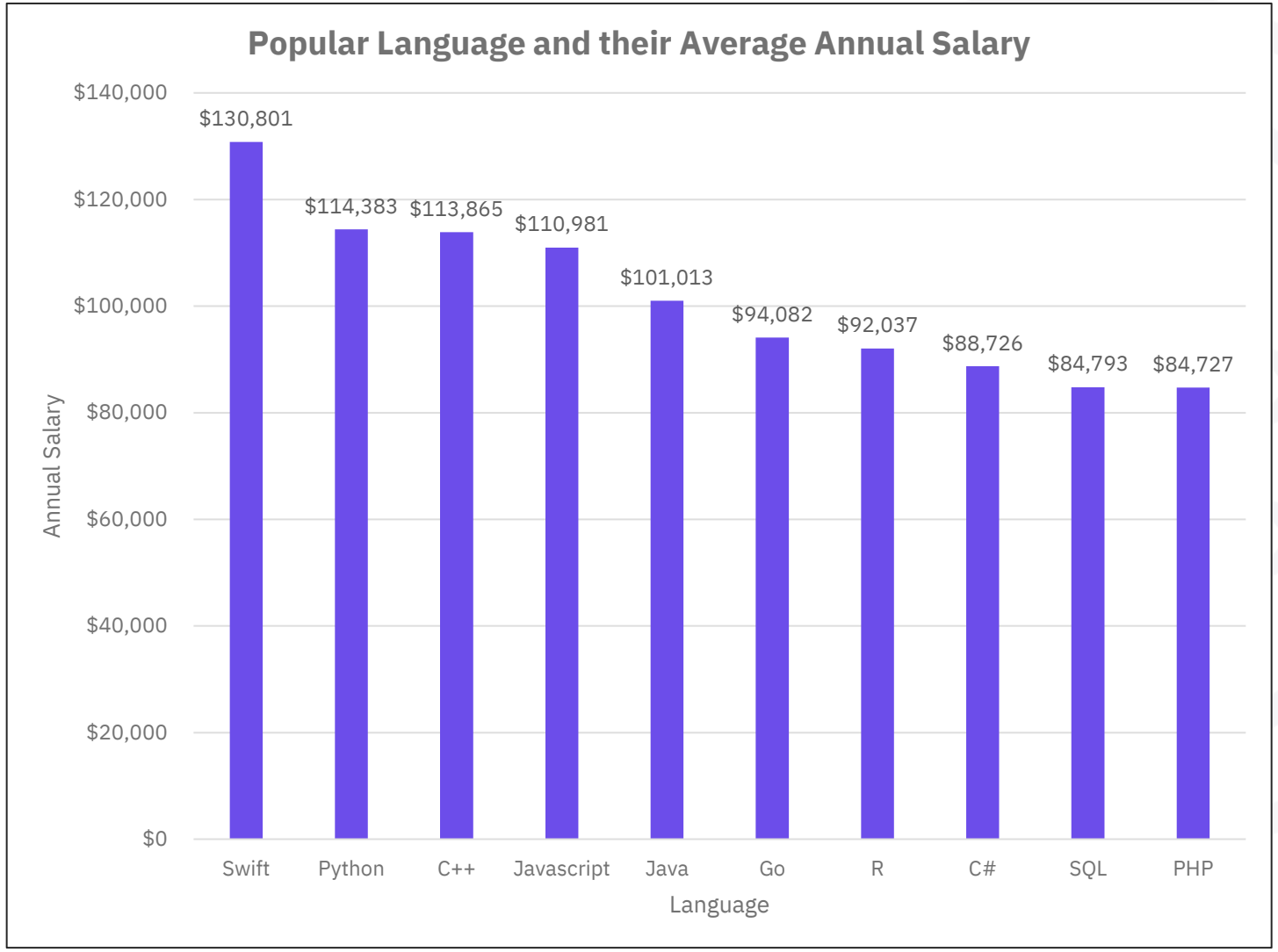
1. C dominates job postings by a very large margin (~25,114)
- ❑ Far exceeds all other technologies
- ❑ Indicates strong demand in systems, embedded, and legacy environments

2. Significant drop after top 1 (very skewed demand distribution)
- ❑ Java (~3,428) and JavaScript (~2,248) far behind
- ❑ Market demand is not evenly distributed

3. Python demand is moderate (~1,173), not top-tier in postings
- ❑ Lower than expected compared to its popularity
- ❑ Suggests Python roles are more specialized (data, AI) rather than mass hiring

4. Databases (MySQL, Oracle) still appear in demand
- ❑ Confirms backend/data infrastructure importance

POPULAR LANGUAGES



1. Swift has the highest salary (~$130K)
❑ Highest-paying but not widely demanded
❑ Niche but high-value skill (iOS ecosystem)

2. Python and C++ are among top-paying (~$113–114K)
❑ Combine high salary + strong demand relevance
❑ Very strategic skills

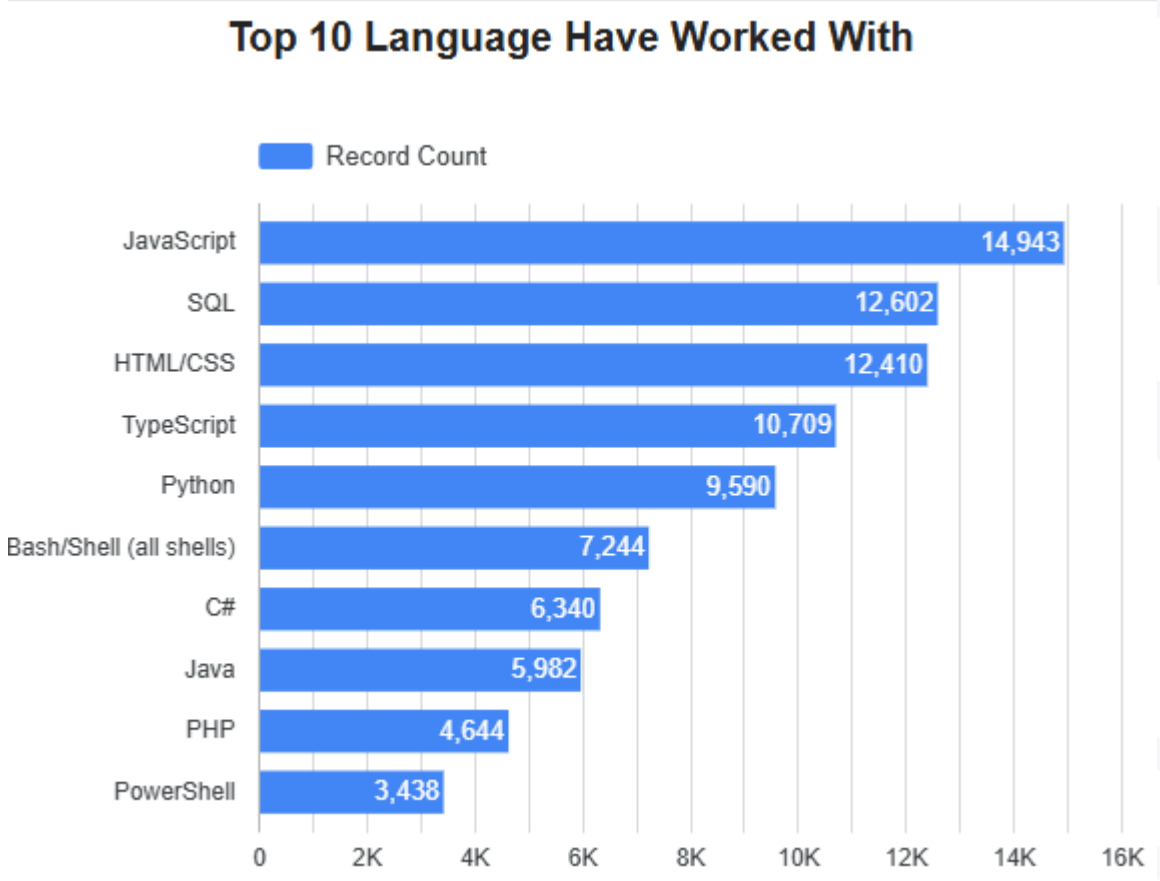
3. JavaScript salary is competitive (~$110K)
❑ Strong balance between demand + pay

4. SQL, PHP have lowest salaries (~$84K range)
❑ Indicates these are more baseline or commodity skills

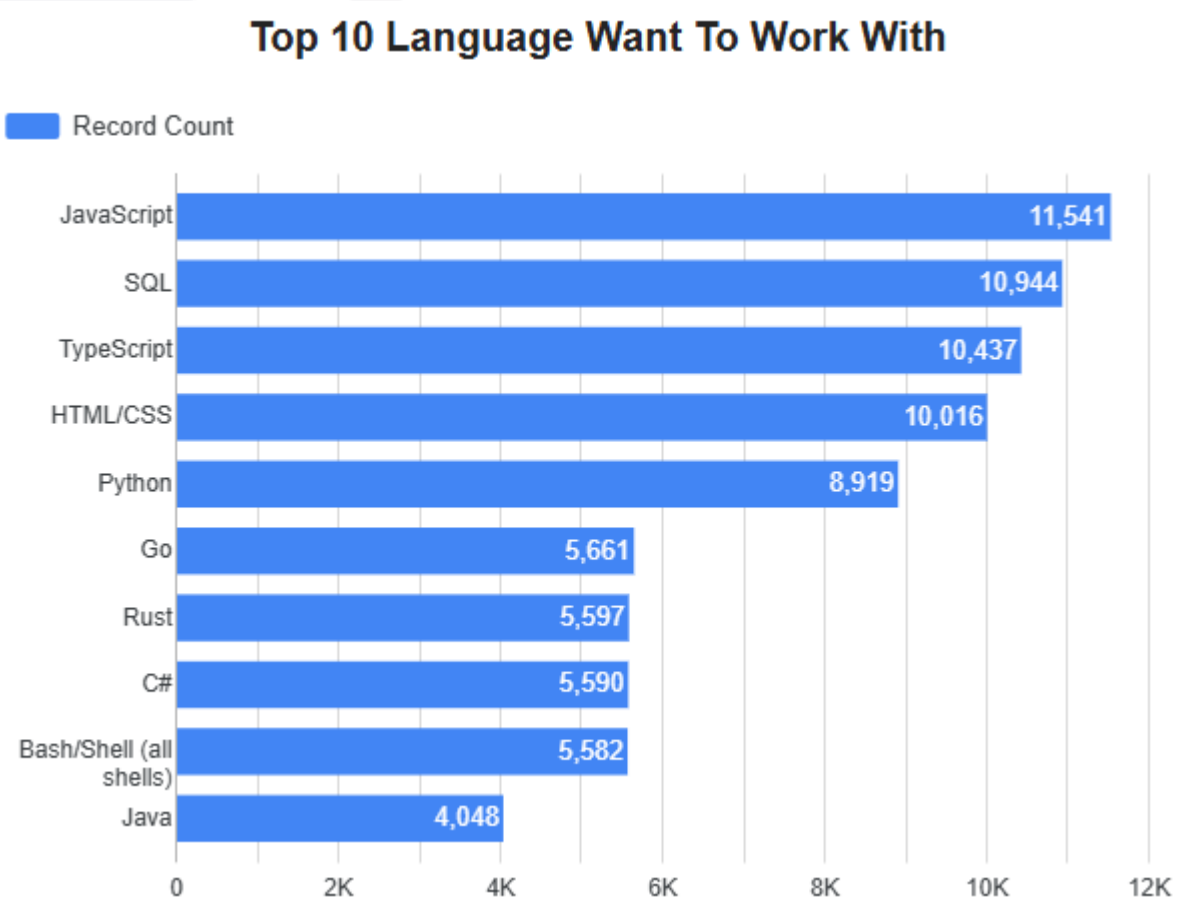
There is a clear mismatch between job demand and salary: high-demand technologies (e.g. C, Java) do not necessarily offer the highest salaries, while niche or specialized languages (e.g. Swift) command premium compensation.

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

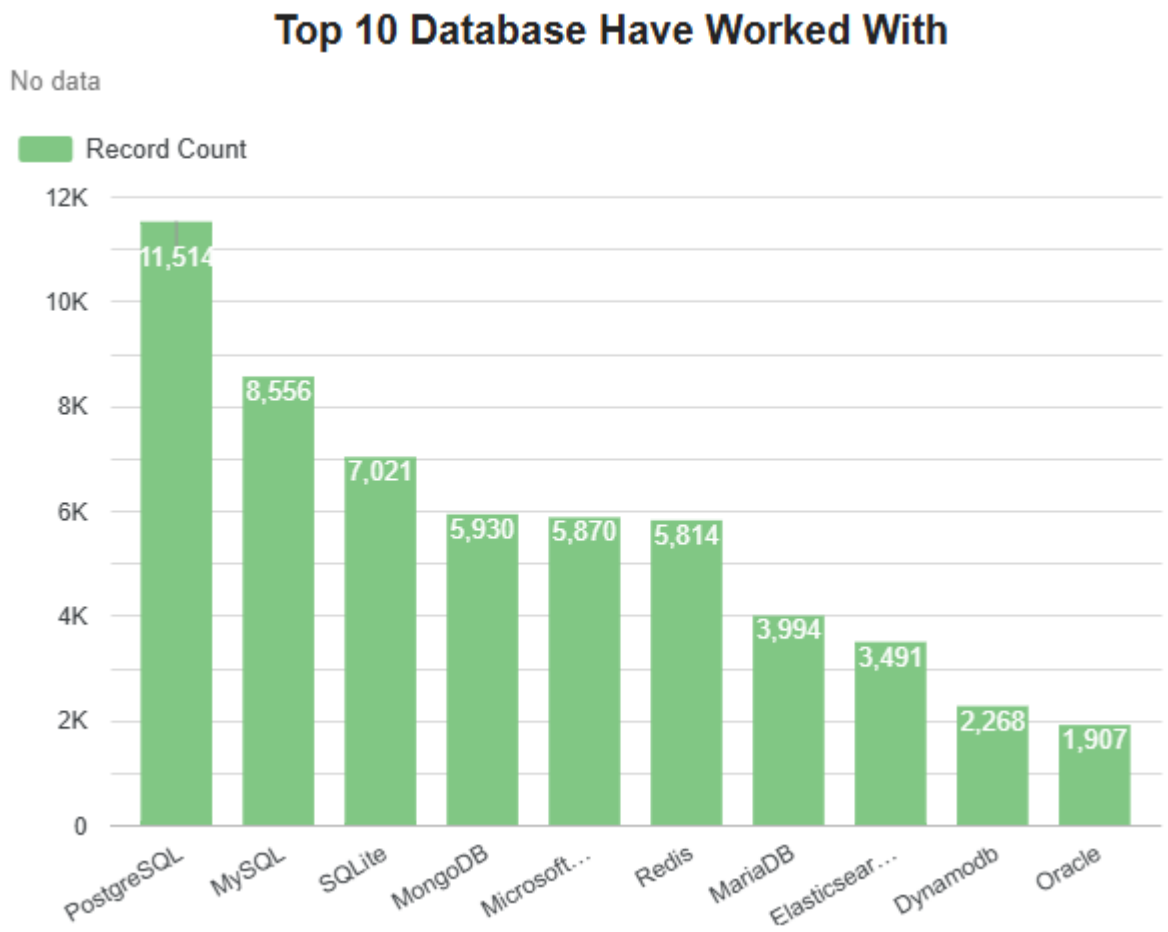
- JavaScript remains the most dominant language (current & future)
- SQL and HTML/CSS continue as foundational technologies
- Python maintains strong, consistent demand
- Go and Rust emerge as high-interest future technologies

Implications

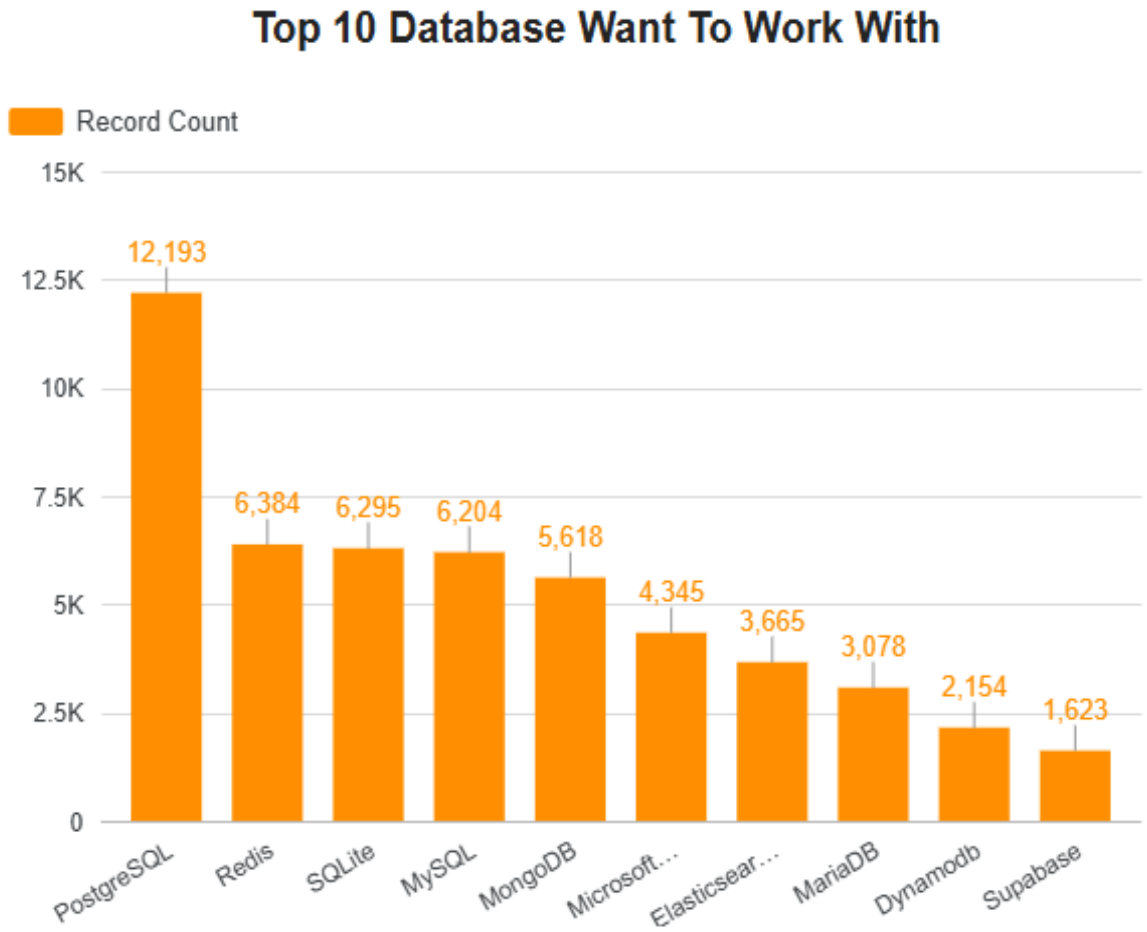
- Web development skills remain highly valuable
- SQL is essential for all data-related roles
- Python is critical for analytics, AI, and automation
- Learning modern languages improves long-term career adaptability

DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Relational databases remain the industry standard
- PostgreSQL shows strong growth trend
- Redis shows stronger forward interest, indicating growing relevance in modern system design

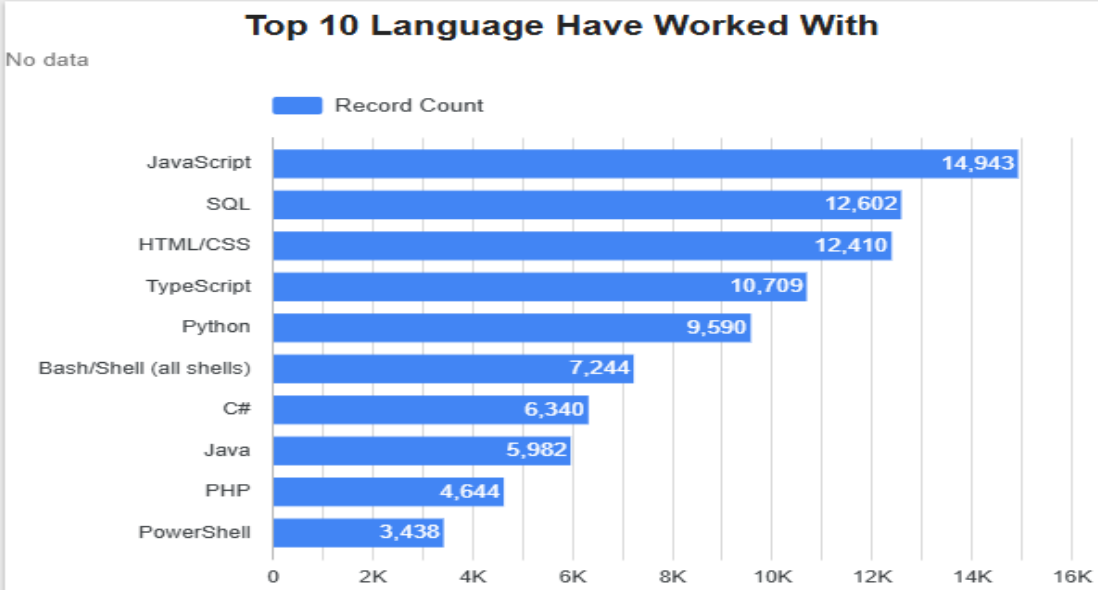
Implications

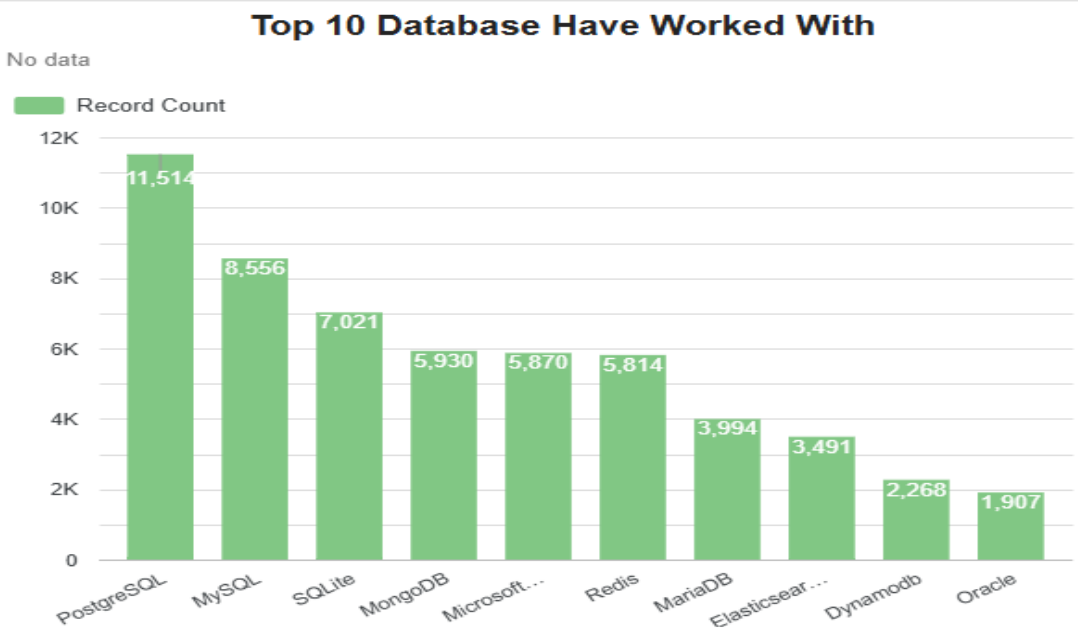
- SQL remains a core technical skill
- PostgreSQL is a valuable database to learn
- Knowledge of Redis can be useful for modern, high-performance and real-time application environments

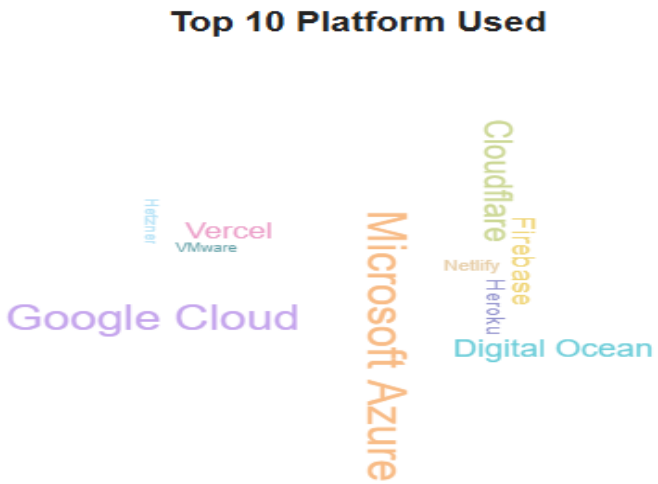
DASHBOARD

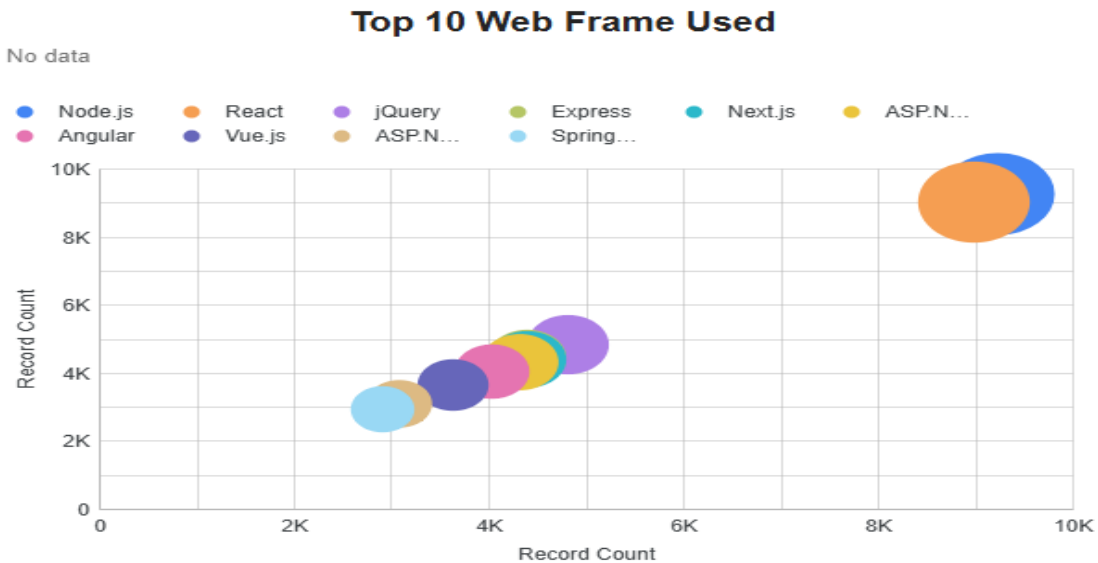
The following dashboard screenshots provide an overview of current technology usage, projected future trends and respondents' demographics

DASHBOARD TAB 1: Current Technology Usage

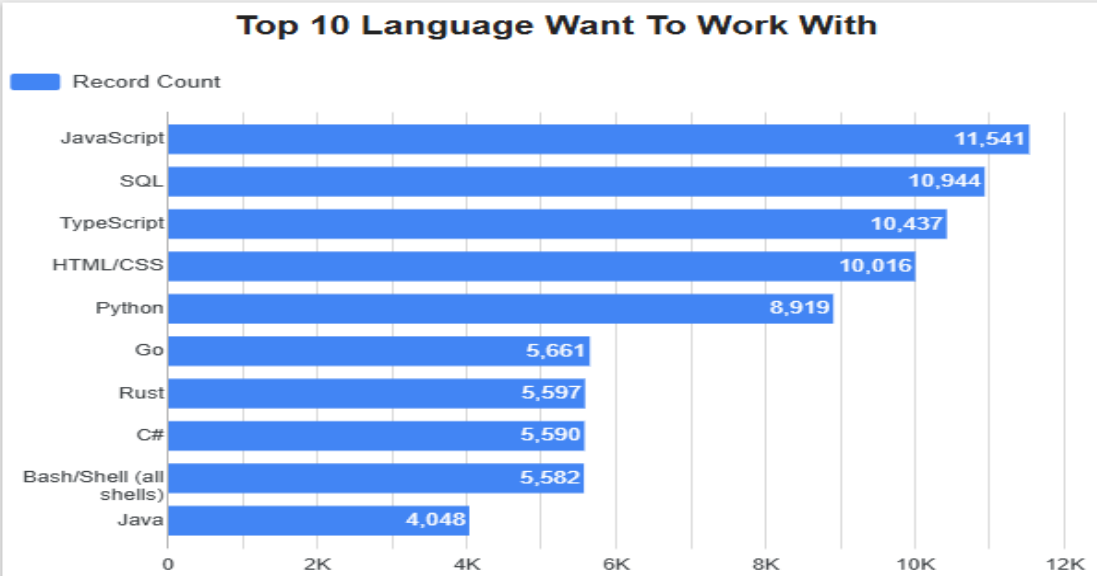


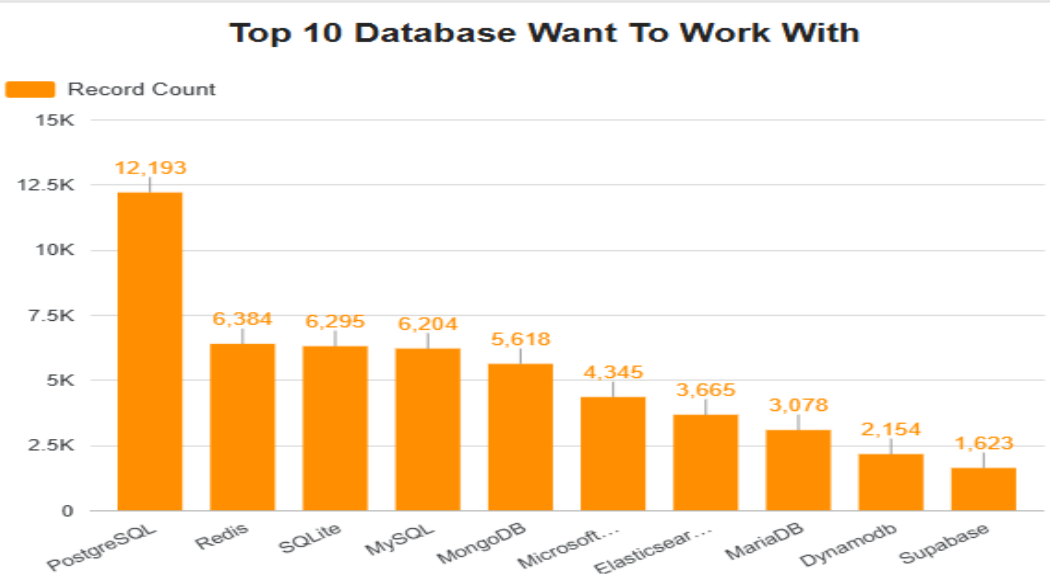


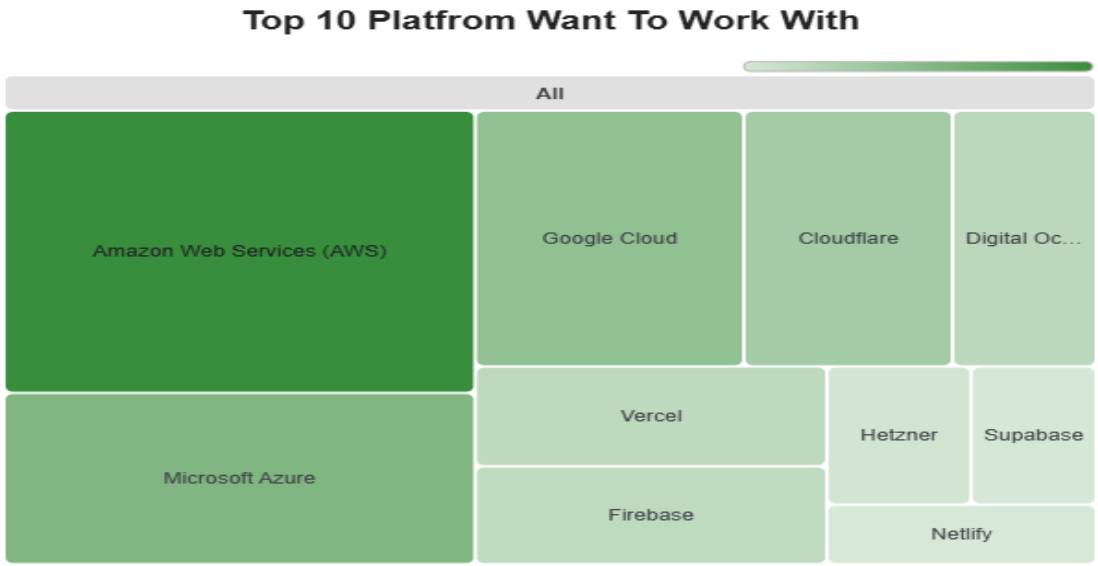


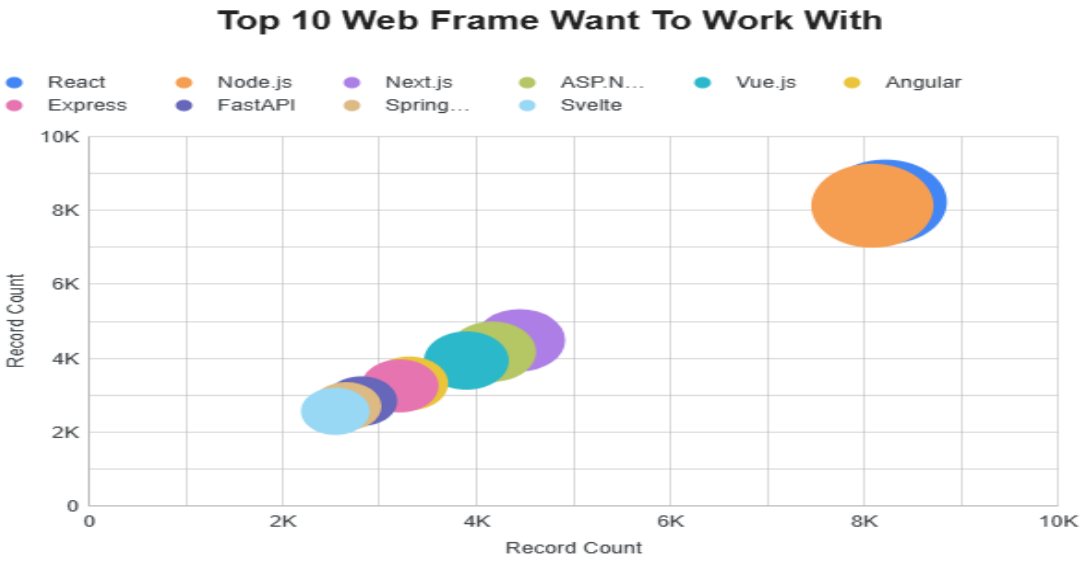


DASHBOARD TAB 2: Future Technology Trends



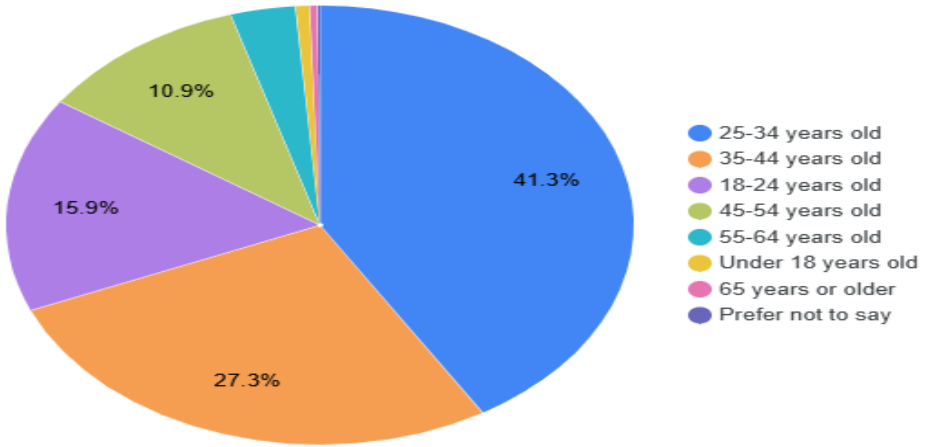




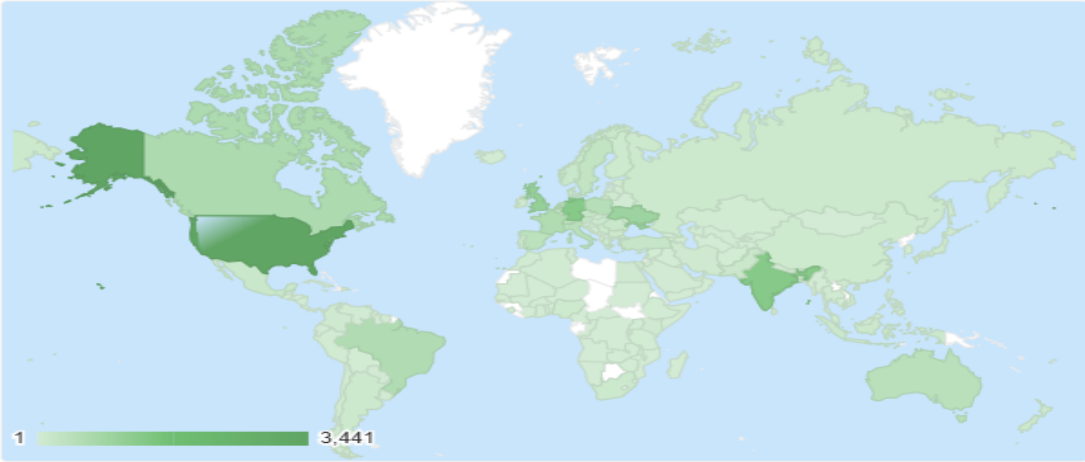


DASHBOARD TAB 3: Demographics

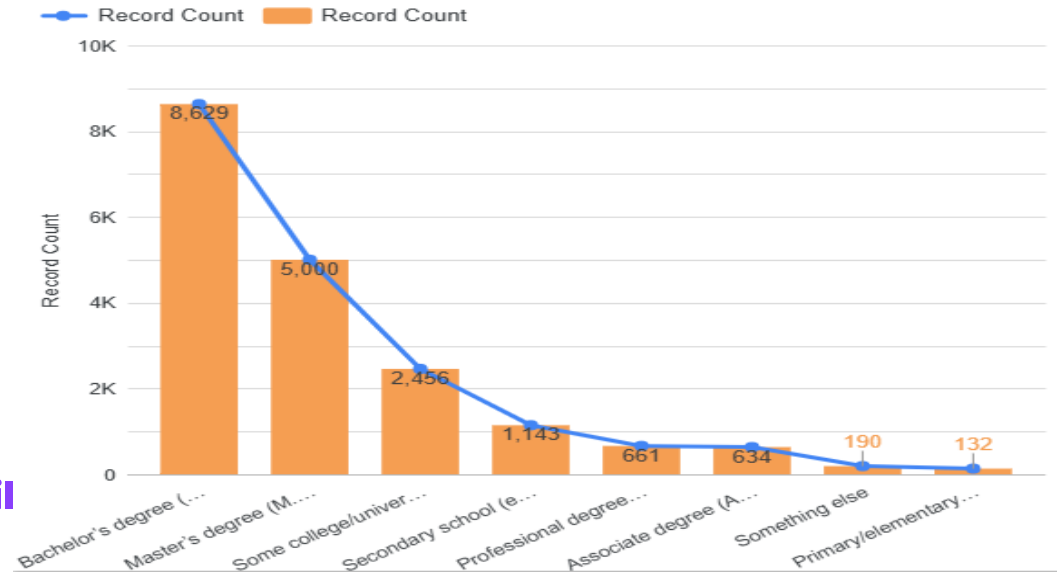
Respondents Distribution by Age



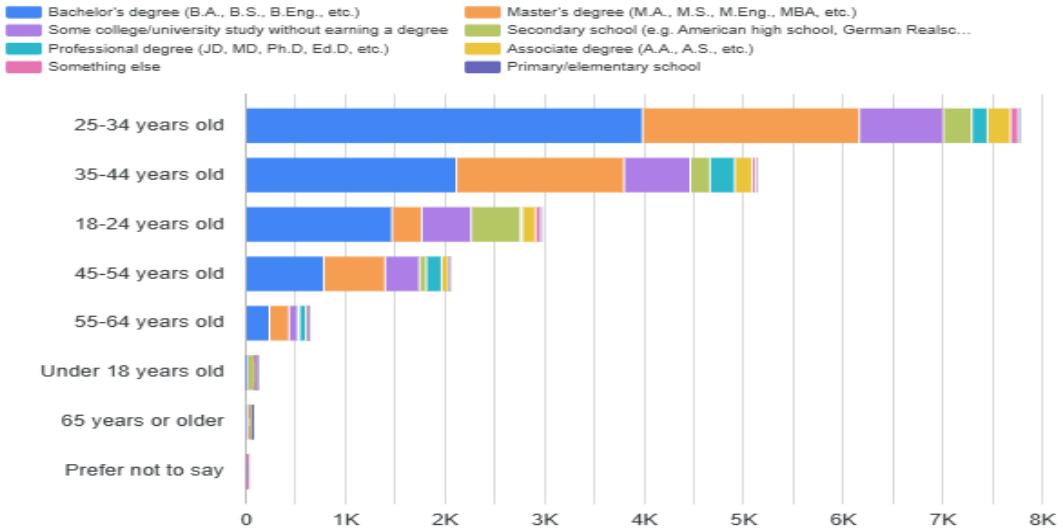
Respondents Count by Country



Distribution of Respondents by Education Level



Record Count by Age and EdLevel



Skil

IBM

DISCUSSION

Programming Languages
- JavaScript, SQL, and Python consistently dominate across both current usage and future interest
- Indicates strong stability in core technology stack
- Emerging languages (Go, Rust) suggest gradual evolution rather than disruption

Insight:
The language ecosystem is mature and stable, with incremental innovation rather than major shifts

Databases
- PostgreSQL leads clearly in both current and future trends
- Relational databases remain dominant across all use cases
- Interest distribution across databases is relatively balanced (no extreme shifts)

Insight:
Database landscape shows high stability, with PostgreSQL emerging as the preferred modern standard

Platforms
- AWS is the most dominant platform, followed by Azure and Google Cloud
- Strong concentration in top 3 providers
- Other platforms (Firebase, Vercel, etc.) serve niche use cases

Insight:
Cloud market is highly consolidated, with hyperscalers controlling the majority of adoption

Web Frameworks
- React and Node.js dominate both current usage and future demand
- Framework adoption is clustered (many options, but similar adoption levels)
- Backend + frontend frameworks both strongly represented

Insight:
Web development ecosystem is diverse but centered around a few dominant frameworks

Overall, the technology landscape is characterized by stable core technologies combined with gradual adoption of emerging tools, indicating a mature yet evolving ecosystem.

OVERALL FINDINGS & IMPLICATIONS (MARKET TREND)

Findings

- Core technologies (JavaScript, Python, SQL) show stable and sustained demand across both current and future trends
- Technology landscape demonstrates gradual evolution rather than disruption
- Modern tools (e.g. Go, Rust, cloud-native technologies) are gaining interest but have not replaced core technologies

Implications

- The technology ecosystem is mature, with strong reliance on established tools and incremental adoption of emerging technologies

OVERALL FINDINGS & IMPLICATIONS (MARKET DYNAMIC & ALIGNMENT)

Findings

- Job postings data shows uneven demand distribution, with certain technologies (e.g. C) dominating specific sectors
- Salary analysis reveals mismatch between demand and compensation
- Survey trends and job market demand show strong alignment for core skills (Python, SQL)

Implications

- Market demand, compensation, and user preference do not always align, highlighting the importance of strategic skill selection

OVERALL FINDINGS & IMPLICATIONS (SKILL STRATEGY & CAREER PATH)

Findings

- Core technologies (Python, SQL, JavaScript) provide broad applicability across multiple roles
- High-paying technologies (e.g. Swift, C++) are often more specialized and niche
- Emerging tools (Go, cloud platforms) show growing interest but lower current adoption
- Data-related roles consistently require a combination of programming + database skills

Implications

- Learners should prioritize high-utility skills (Python + SQL) for maximum employability
- Career growth can be accelerated by adding specialized or niche skills for higher compensation
- A balanced skill set (core + emerging tools) provides both short-term job access and long-term career flexibility
- Building real-world projects is essential to demonstrate applied skill integration

OVERALL FINDINGS & IMPLICATIONS (FUTURE OUTLOOK)

Findings

- Core technologies (JavaScript, Python, SQL) are expected to remain dominant in the near future
- Increasing interest in modern, high-performance languages (e.g. Go, Rust)
- Strong growth in cloud platforms (AWS, Azure, Google Cloud) indicates shift toward cloud-native development
- Expanding use of real-time and scalable systems (e.g. Redis, distributed technologies)

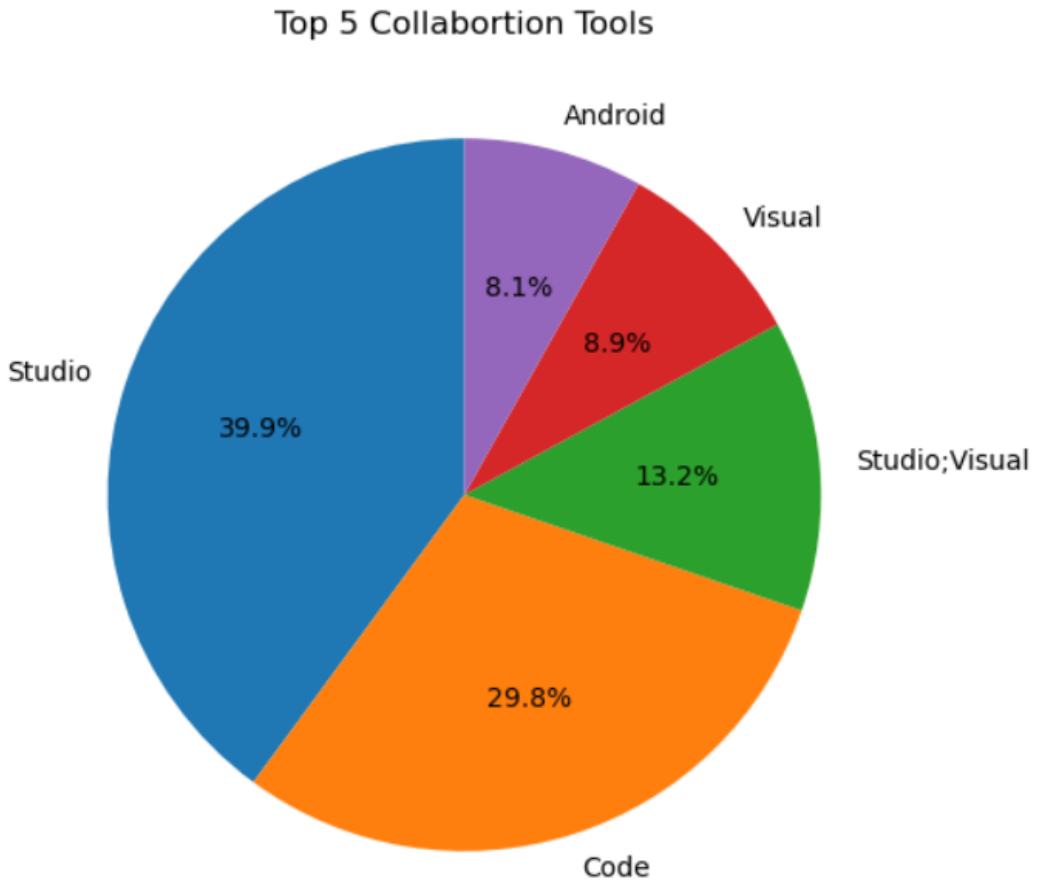
Implications

- Future tech landscape will be hybrid, combining stable core technologies with emerging tools
- Professionals should develop cloud and scalability-related skills to stay competitive
- Understanding system design, performance, and distributed systems will become increasingly valuable
- Continuous upskilling is critical as technology evolution accelerates

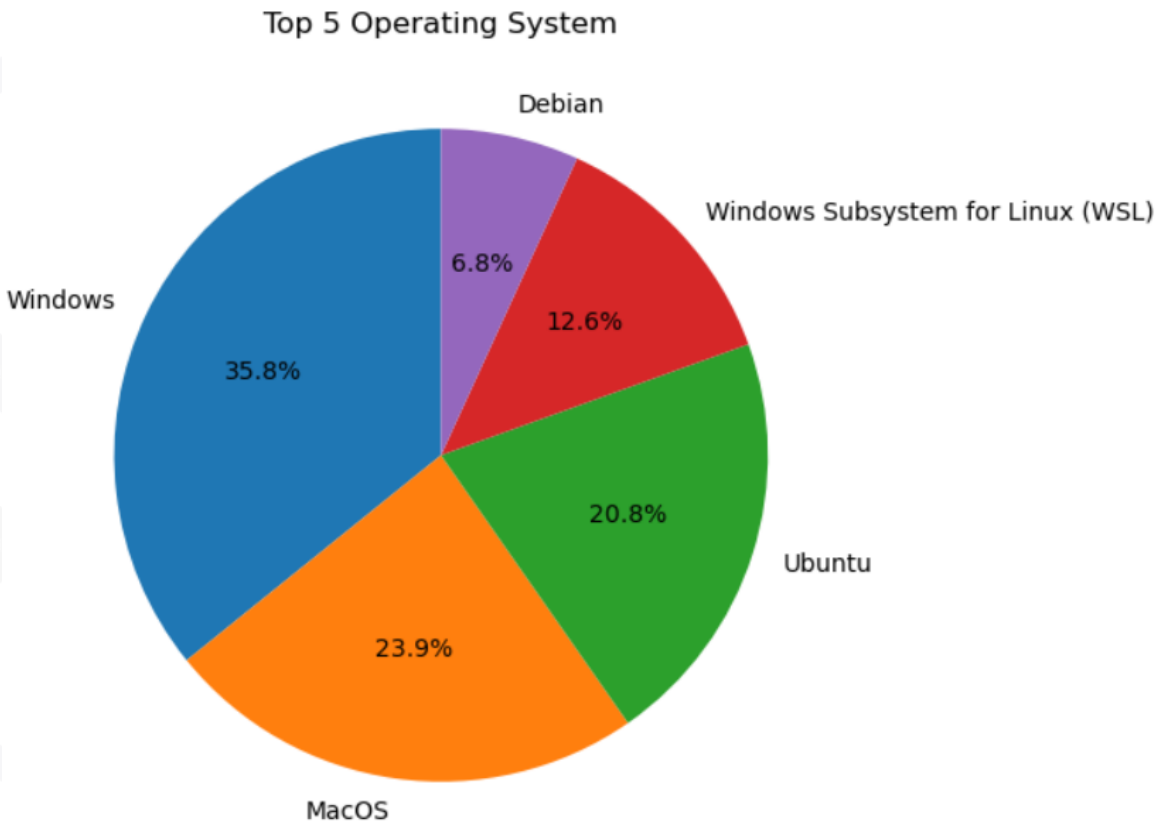
CONCLUSION

- The technology landscape is stable yet evolving, with core technologies (JavaScript, Python, SQL) remaining dominant while new tools are gradually adopted
- Market dynamics show that demand, salary, and user preference do not always align, requiring careful and strategic skill selection
- An effective career path requires balancing foundational skills for employability with specialized skills for differentiation and growth
- Looking ahead, the industry will move toward a hybrid ecosystem, combining core technologies with cloud, scalability, and modern system design

APPENDIX


Top 5 Collabortion Tools
Android
Visual
Studio;Visual
Code
Studio
8.1%
8.9%
13.2%
29.8%
39.9%

APPENDIX


Top 5 Operating System
Debian
Windows Subsystem for Linux (WSL)
Windows
Ubuntu
MacOS
6.8%
12.6%
35.8%
20.8%
23.9%

COLLABORATION TOOLS

Findings

- Visual Studio–related tools (Visual Studio, VS Code, combinations) dominate usage (~70% combined)
- VS Code alone represents a significant share (~29.8%), indicating strong adoption of lightweight development environments
- Traditional IDE usage (e.g. Visual Studio) remains relevant but is no longer the only dominant tool
- Other tools (e.g. Android Studio) have relatively niche usage

Implications

- Development workflows are shifting toward flexible, lightweight, and customizable tools
- VS Code's popularity suggests strong preference for extensible, cross-platform environments
- Developers increasingly adopt multiple tools simultaneously, rather than relying on a single IDE
- Tool ecosystems that support plugins and integrations are becoming more valuable

OPERATING SYSTEM

Findings

- Windows remains the most widely used operating system (~35.8%)
- macOS (~23.9%) and Ubuntu (~20.8%) together represent a significant developer preference for Unix-based systems
- Windows Subsystem for Linux (WSL) (~12.6%) shows growing adoption of hybrid environments
- Traditional Linux distributions (e.g. Debian) have relatively smaller standalone usage

Implications

- Development environments are shifting toward Unix/Linux-based ecosystems, even among Windows users
- Tools and workflows that support cross-platform or Linux compatibility are increasingly important
- The rise of WSL suggests developers prefer flexibility rather than switching fully between operating systems
- This additional analysis highlights a shift toward hybrid and Unix-based development environments, providing deeper insight beyond the core dataset findings.

Sources

- IBM Skills Network. Popular Programming Languages. Retrieved April 2026 from: https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DA0321EN-SkillsNetwork/labs/datasets/ProgrammingLanguages.html
- Stack Overflow. (2019). *Annual Developer Survey 2019*. Retrieved April 2026 from: https://insights.stackoverflow.com/survey.